

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 2 – Architecture Design Assignment

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Architecture Design Assignment
Weightage:	20%	Total Marks:	25
CO2 Statement:	Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills		
Student Name:	G. Bharathi	Reg. No.:	192371079
Date:	13/09/2020	Duration:	60 minutes

Code of Conduct

I G. Bharathi (192371079) (Name / Reg No) certify that this submission is my original work and that I have adhered to the provision of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Architecture Design Assignment

Instructions to Students:

Each student will be assigned an **individual software project problem statement**. Based on the assigned problem, analyze the system requirements and design an appropriate software architecture by applying software engineering principles. Your submission should demonstrate your ability to select, justify, and evaluate a suitable architectural style.

Question

Based on your **assigned problem statement**, perform an **Architecture Design Analysis** and prepare an architecture design report by answering the following questions:

1. Analyze System Requirements

- Study the given problem statement and identify the system objectives.
- Analyze the functional and non-functional requirements that influence the software architecture.
- Identify key quality attributes such as scalability, maintainability, reliability, availability, performance, and security.

2. Select a Suitable Software Architecture

- Select an appropriate architectural style for the proposed system (e.g., Layered Architecture, Client-Server, MVC, Microservices, Event-Driven, Service-Oriented Architecture, Serverless, or Hybrid Architecture).
- Justify why the selected architecture is the most suitable for the given problem.

3. Draw the Software Architecture Diagram

- Design a clear architecture diagram illustrating the overall structure of the proposed system.
- Include all major layers/components, databases, external systems, APIs, and communication mechanisms.
- Use appropriate software architecture notations and labels.

4. Explain Components and Their Interactions

- Describe the purpose and responsibilities of each architectural component.
- Explain how the components communicate and exchange data.
- Illustrate the overall workflow of the system.

5. Discuss Quality Attributes

Evaluate how the proposed architecture addresses the following aspects:

- Scalability
- Security
- Performance
- Reliability
- Maintainability
- Availability

6. Justify Architectural Decisions

- Explain the rationale behind your architectural choices.
- Discuss the trade-offs considered while selecting the architecture.
- Explain how the chosen architecture supports future enhancements, system integration, and long-term maintainability.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
System Requirement Analysis (30%)	Thoroughly analyzes the problem statement, accurately identifies functional and non-functional requirements, and clearly explains quality attributes influencing the architecture.	Analyzes most requirements with minor omissions or limited explanation.	Identifies only basic requirements with partial analysis.	Requirement analysis is incomplete, inaccurate, or lacks clarity.
Selection of Software Architecture (25%)	Selects the most appropriate architectural style with strong technical justification aligned to system requirements.	Selects a suitable architecture with reasonable justification.	Architecture is partially suitable with limited justification.	Inappropriate architecture selected or justification is missing.
Architecture Diagram and Component Design (20%)	Produces a clear, complete, and well-structured architecture diagram with all	Diagram is mostly complete with minor omissions or notation issues.	Diagram includes only essential components and	Diagram is incomplete, unclear, or

	major components, interfaces, and interactions accurately represented.		lacks sufficient detail.	technically incorrect.
Component Interaction and Quality Attribute Analysis (15%)	Clearly explains component responsibilities, interactions, and thoroughly discusses scalability, security, performance, reliability, maintainability, and availability.	Explains most components and quality attributes with minor gaps.	Provides limited explanation of interactions and addresses only some quality attributes.	Component interactions and quality attributes are poorly explained or missing.
Architectural Justification and Technical Presentation (10%)	Provides strong justification for architectural decisions, discusses trade-offs and future enhancements, and presents a well-organized, professional report.	Provides adequate justification with minor gaps; report is well organized.	Limited justification with minimal discussion of trade-offs or future scope; report needs improvement.	Justification is weak or absent; report lacks organization and technical clarity.

Criteria	Mark
System Requirement Analysis (30%)	/5
Selection of Software Architecture (25%)	/5
Architecture Diagram and Component Design (20%)	/5
Component Interaction and Quality Attribute Analysis (15%)	/5
Architectural Justification and Technical Presentation (10%)	/5
TOTAL	/25

COLLEGE ADMISSION COUNSELING SYSTEM

Architecture Design Analysis and Report

1. Analyze System Requirements

The system requires secure student access, college/course information, counseling workflows, appointment scheduling, application tracking, document handling, notifications, and administrative reporting. These requirements influence an architecture that separates user interaction, business rules, data access, and persistent storage.

Quality Attributes

- Scalability
- Security
- Performance
- Reliability
- Maintainability
- Availability
- Usability
- Privacy

2. Select a Suitable Software Architecture

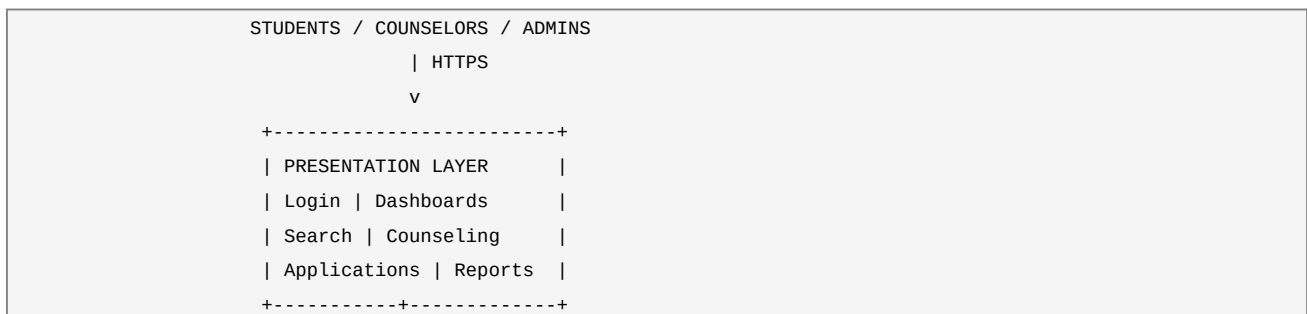
Selected Architecture: Layered Architecture

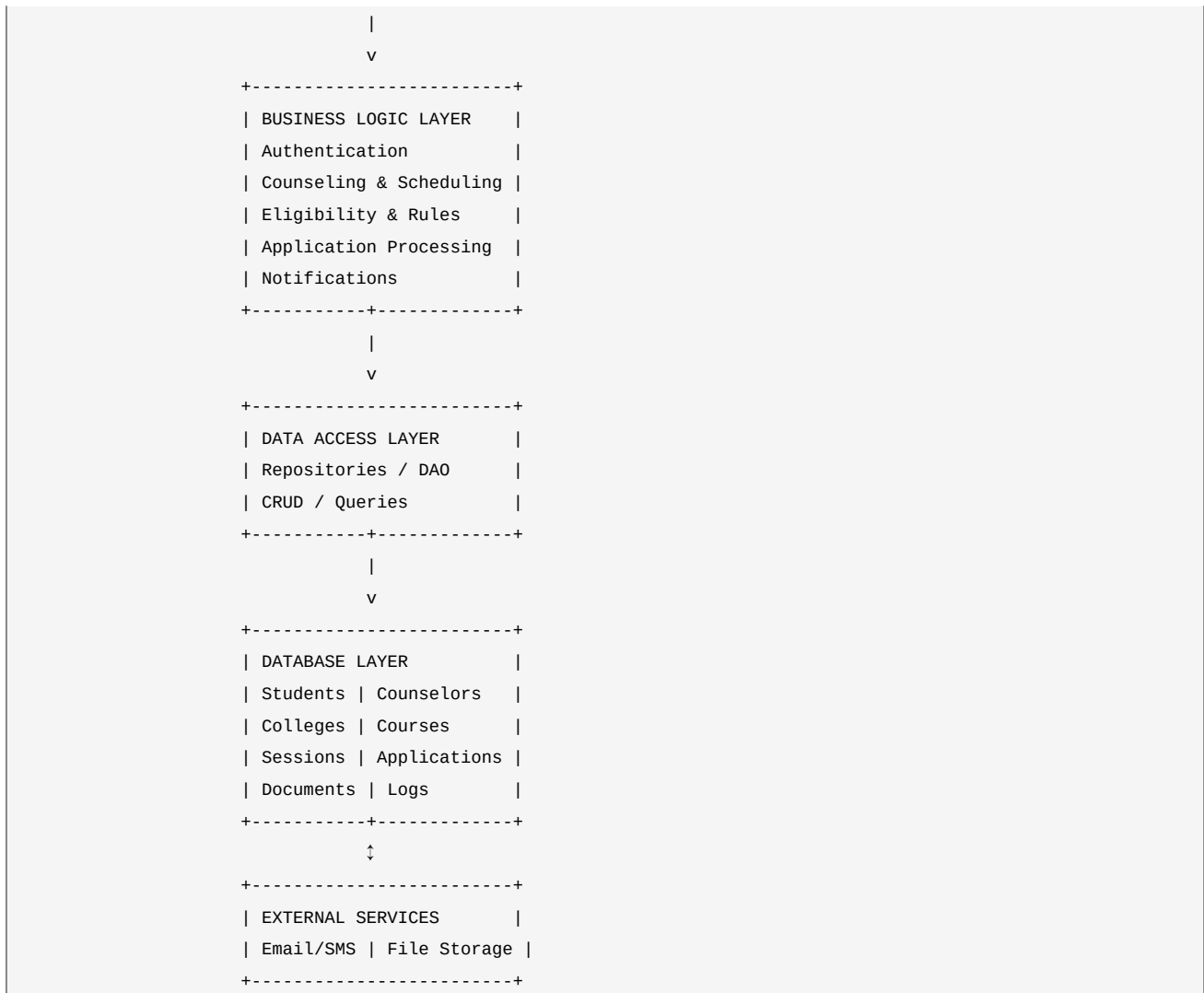
A four-layer architecture is selected: Presentation Layer, Business Logic Layer, Data Access Layer, and Database Layer. External services such as notifications and document storage communicate through secure APIs. This structure follows the assignment's requirement to identify layers/components, databases, external systems, APIs, and communication mechanisms.

Why Suitable

- Separates UI from counseling and admission rules
- Protects database access
- Supports independent testing and maintenance
- Makes college/course and application modules easier to extend
- Allows external notification/document services
- Provides a clear structure for future mobile or web clients

3. Software Architecture Diagram





4. Explain Components and Their Interactions

Presentation Layer

Provides student, counselor and administrator interfaces. It accepts input, displays search results, appointments, recommendations and application status, and sends requests to business services.

Business Logic Layer

Handles authentication, authorization, eligibility rules, counseling scheduling, recommendations, application processing, document validation, notifications and reporting.

Data Access Layer

Provides repositories/DAO components for storing and retrieving students, colleges, courses, appointments, applications, documents and reports.

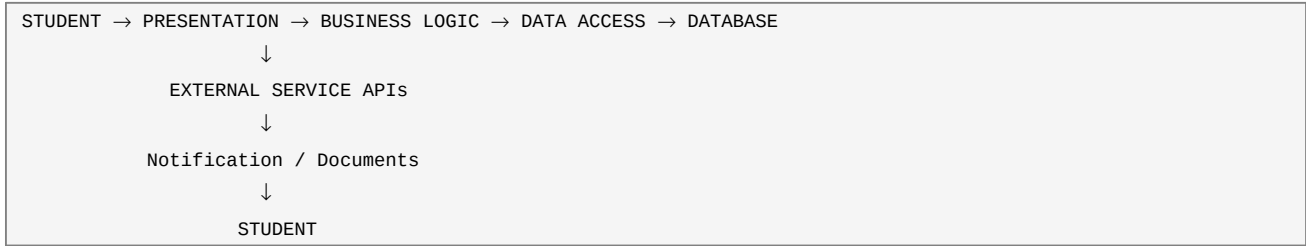
Database Layer

Stores structured system data and status history. Access is controlled through the data access layer.

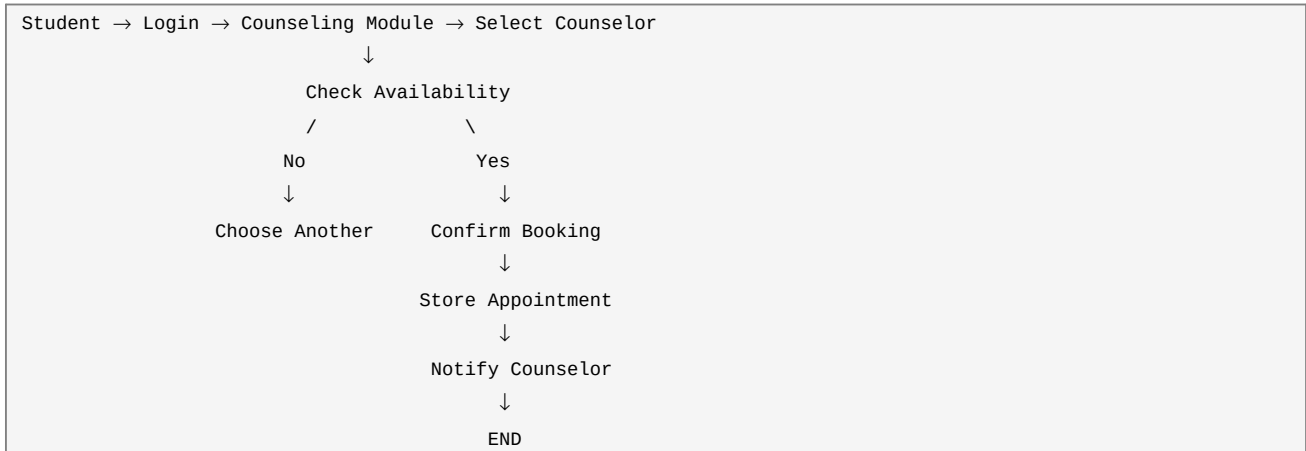
External Services

Email/SMS service sends reminders and application updates. File storage handles uploaded admission documents. APIs provide secure communication.

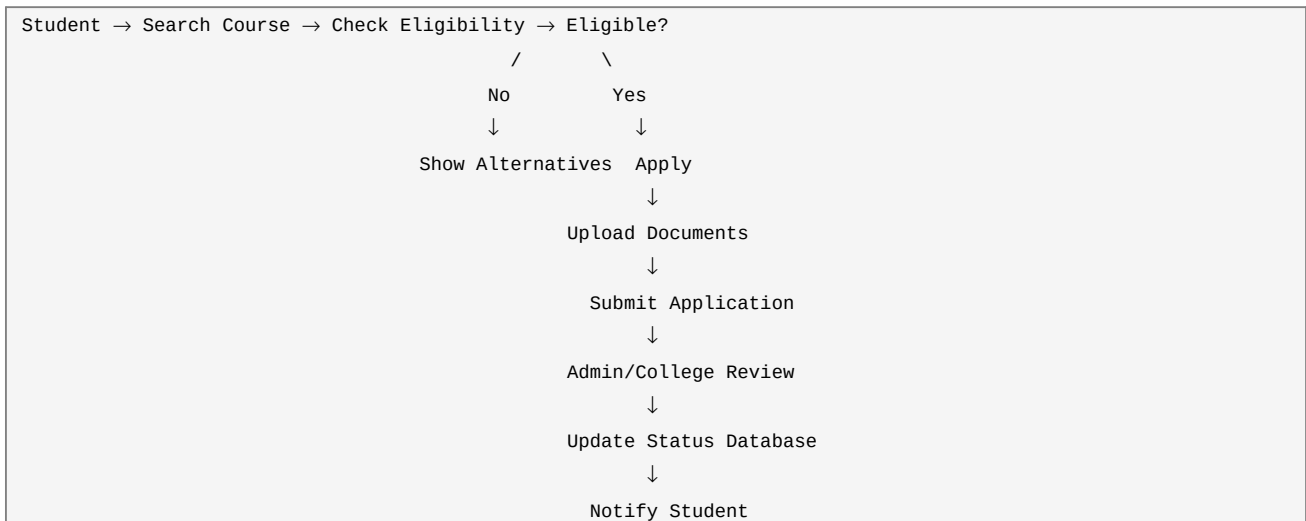
5. Overall Workflow



6. Counseling Interaction Flowchart



7. Application Processing Flowchart



8. Discuss Quality Attributes

Quality Attribute	Architecture Support
Scalability	Layer separation allows growth of users, colleges, applications and counseling records; caching/indexing can improve scalability.
Security	Role-based authorization, HTTPS, validation, protected document access and controlled database connectivity.
Performance	Indexed searches, optimized queries, caching and efficient API calls improve response time.
Reliability	Validation, transaction handling, backups, logging and status history preserve important records.
Maintainability	Independent layers and modules simplify testing, debugging and enhancement.

Availability	Backups, monitoring, recovery procedures and reliable service deployment improve availability.
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9. Justify Architectural Decisions

Layered Architecture is appropriate because the College Admission Counseling System has distinct responsibilities: user interaction, admission/counseling rules, data operations and persistence. It reduces coupling, improves security, supports testing and provides a clear path for future integration with mobile applications, college data services, notification platforms and document storage.

10. Architectural Trade-Offs

Decision	Advantage	Trade-Off
Layered Architecture	Clear separation and maintainability	Some additional processing/abstraction
Central Database	Consistent student/application records	Potential bottleneck at very high scale
External Notification API	Fast integration	Depends on third-party service
Document Storage	Scalable file handling	Requires access control and storage management

11. Future Enhancements and Long-Term Maintainability

- Mobile application for students and counselors
- Integration with college/university admission APIs
- AI-assisted counseling recommendations as an optional future module
- Multilingual support
- Advanced analytics and dashboards
- Automated deadline reminders
- Scalable cloud deployment
- Independent services for high-volume modules if future scale requires them

12. Conclusion

The proposed Layered Architecture provides a clear, secure and maintainable foundation for the College Admission Counseling System. It separates presentation, business rules, data access and storage while supporting counseling, college search, eligibility, applications, documents and notifications. The structure addresses the quality attributes and architectural requirements specified in the assignment.